

2.10.43

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Question

Find the area of the quadrilateral ABCD, whose vertices are A(3, 1) , B (2, 4) , C (4, 1) and D (3, 4).

Theoretical Solution

Given the points,

Name	Point
A	$\begin{pmatrix} -3 \\ -1 \end{pmatrix}$
B	$\begin{pmatrix} -2 \\ -4 \end{pmatrix}$
C	$\begin{pmatrix} 4 \\ -1 \end{pmatrix}$
D	$\begin{pmatrix} 3 \\ 4 \end{pmatrix}$

Table: Vectors of the vertices of quadrilateral

We can say that,

$$\text{Area}(ABCD) = \text{Area}(ABC) + \text{Area}(ADC) \quad (1)$$

$$\text{Area}(\triangle ABC) = \frac{1}{2} \|(\mathbf{B} - \mathbf{A}) \times (\mathbf{B} - \mathbf{C})\| \quad (2)$$

$$\text{Area}(\triangle ADC) = \frac{1}{2} \|(\mathbf{D} - \mathbf{A}) \times (\mathbf{D} - \mathbf{C})\| \quad (3)$$

$$(\mathbf{B} - \mathbf{A}) = \begin{pmatrix} -2 \\ -4 \end{pmatrix} - \begin{pmatrix} -3 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \end{pmatrix} \quad (4)$$

$$(\mathbf{B} - \mathbf{C}) = \begin{pmatrix} -2 \\ -4 \end{pmatrix} - \begin{pmatrix} 4 \\ -1 \end{pmatrix} = \begin{pmatrix} -6 \\ -3 \end{pmatrix} \quad (5)$$

$$\|(\mathbf{B} - \mathbf{A}) \times (\mathbf{B} - \mathbf{C})\| = \left\| \begin{pmatrix} 1 \\ -3 \end{pmatrix} \times \begin{pmatrix} -6 \\ -3 \end{pmatrix} \right\| \quad (6)$$

$$\implies |-3 - (-6)(-3)| = |-3 - 18| = 21 \quad (7)$$

and

$$(\mathbf{D} - \mathbf{A}) = \begin{pmatrix} 3 \\ 4 \end{pmatrix} - \begin{pmatrix} -3 \\ -1 \end{pmatrix} = \begin{pmatrix} 6 \\ 5 \end{pmatrix} \quad (8)$$

$$(\mathbf{D} - \mathbf{C}) = \begin{pmatrix} 3 \\ 4 \end{pmatrix} - \begin{pmatrix} 4 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 5 \end{pmatrix} \quad (9)$$

$$\|(\mathbf{D} - \mathbf{A}) \times (\mathbf{D} - \mathbf{C})\| = \left\| \begin{pmatrix} 6 \\ 5 \end{pmatrix} \times \begin{pmatrix} -1 \\ 5 \end{pmatrix} \right\| \quad (10)$$

$$\implies |(6)(5) - 5(-1)| = 30 + 5 = 35 \quad (11)$$

$$\text{Area}(ABCD) = \frac{1}{2} |(-21)| + \frac{1}{2} |(35)| = \frac{56}{2} = 28 \text{sq.units} \quad (12)$$

$\therefore$ The Area of the quadrilateral $ABCD$ is 28 sq.units.

```
#include <stdio.h>
#include <math.h>

// Function to calculate area of quadrilateral using shoelace
// formula
float quadArea(int x[], int y[], int n) {
    float area = 0.0;
    int i, j;

    for (i = 0; i < n; i++) {
        j = (i + 1) % n; // Next vertex index
        area += (x[i] * y[j]) - (y[i] * x[j]);
    }
}
```

```
    return fabs(area) / 2.0;
}

// Expose function for Python
float findQuadArea(int x1, int y1, int x2, int y2,
                  int x3, int y3, int x4, int y4) {
    int x[4] = {x1, x2, x3, x4};
    int y[4] = {y1, y2, y3, y4};
    return quadArea(x, y, 4);
}
```

```
import ctypes
import matplotlib.pyplot as plt

# Load compiled shared C library
quad_lib = ctypes.CDLL('./code4.so')

# Define function signature
quad_lib.findQuadArea.argtypes = [ctypes.c_int, ctypes.c_int,
                                   ctypes.c_int, ctypes.c_int,
                                   ctypes.c_int, ctypes.c_int,
                                   ctypes.c_int, ctypes.c_int]
quad_lib.findQuadArea.restype = ctypes.c_float
```

```
# Take input from user
print(Enter coordinates of quadrilateral ABCD:)

x1, y1 = map(int, input(Enter A (x y): ).split())
x2, y2 = map(int, input(Enter B (x y): ).split())
x3, y3 = map(int, input(Enter C (x y): ).split())
x4, y4 = map(int, input(Enter D (x y): ).split())

# Call C function
area = quad_lib.findQuadArea(x1, y1, x2, y2, x3, y3, x4, y4)
```

```
# Store points and labels
points = [(x1, y1), (x2, y2), (x3, y3), (x4, y4)]
labels = [A, B, C, D]

# Close polygon for plotting
x_vals = [p[0] for p in points] + [points[0][0]]
y_vals = [p[1] for p in points] + [points[0][1]]

# Plot quadrilateral
plt.figure(figsize=(7,7))
plt.plot(x_vals, y_vals, 'b-o', linewidth=2)
```

```
# Fill (shade) the quadrilateral
plt.fill(x_vals, y_vals, color=skyblue, alpha=0.5, label=
    Quadrilateral Area)

# Mark vertices with labels + coordinates
for (x, y), label in zip(points, labels):
    plt.text(x+0.2, y+0.2, f{label}{(x,y)}, fontsize=11, color=
        red)

# Draw diagonal AC
A = points[0] # A(x1, y1)
C = points[2] # C(x3, y3)
plt.plot([A[0], C[0]], [A[1], C[1]], 'r--', linewidth=2, label=
    Diagonal AC)
```

```
# Formatting
plt.title(fQuadrilateral ABCD (Area = {area:.2f} sq. units))
plt.axhline(0, color=black, linewidth=0.5)
plt.axvline(0, color=black, linewidth=0.5)
plt.grid(True, linestyle=--, alpha=0.6)
plt.gca().set_aspect('equal', adjustable='box')
plt.legend()

# Save the figure
plt.savefig(fig_4.png, dpi=300, bbox_inches=tight)

# Show the plot
plt.show()
```

